

Machine Learning Assignment

Polynomial Regression

GitHub: https://github.com/praneethem/ML_Assignment

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1 Introduction

This assignment has two problems, and in both I have to predict a number y using only polynomial regression.

- **VAR1 (steam turbine):** six inputs $x_1, \dots, x_6$, and the target is the Net Power Score.
- **VAR2 (thermal reservoir):** three spatial inputs x_1, x_2, x_3 , and the target is the Thermal Anomaly Score.

Each problem has 1000 training rows with y and 1000 test rows without y . Since the model type is fixed, the real work is choosing the polynomial degree (and which features to use) so that the model does not underfit or overfit. My final choices are:

- **VAR1:** all six features, degree 4 (209 polynomial terms).
- **VAR2:** all three features, degree 8 (164 polynomial terms).

2 Looking at the Data

I started by checking the files. None of the four files has missing values, all columns are numeric, and every feature lies between -1 and 1 . The training sets have no duplicate rows. The test sets contain 6 duplicate rows (VAR1) and 7 duplicate rows (VAR2).

Table 1: Summary of the target y in the training data.

	mean	std	min	max
VAR1 y	0.813	3.208	-9.924	12.073
VAR2 y	2.155	6.629	-29.811	40.028

Correlations. In VAR1 no single feature has a strong straight-line relation with y (the largest correlation is -0.32 , for x_5), while x_1 to x_4 are moderately correlated with each other (0.40 to 0.49). In VAR2, x_2 has the strongest link with y (0.51), and x_1 and x_3 are weak (Figure 1).

Shape of the relationship. The scatter plots (Figures 2 and 3) are clearly curved, and in VAR2 the spread of y grows towards the edges of the cube. A plain straight-line fit confirms this: its cross-validated R^2 is only 0.136 (VAR1) and 0.284 (VAR2). Going to degree 2 raises these to 0.692 and 0.547 , so a higher degree is needed.

Train versus test inputs. For VAR2 the test inputs look like the training inputs. For VAR1 they do not: the standard deviation of each feature is about 0.81 to 0.84 in the test set but about 0.71 to 0.72 in training, and 51% of the VAR1 test feature values sit exactly on ± 1 compared with 30% in training. So the VAR1 test points lie closer to the edges of the input range, where the training data is thinner. I come back to this in Section 6.

3 Method

Model. For a chosen degree d , `PolynomialFeatures` builds every term whose powers add up to at most d , including mixed terms such as $x_1 x_2^2$. This is the same definition of degree as in the assignment. With k features the number of terms is

$$p(k, d) = \binom{k+d}{d} - 1,$$

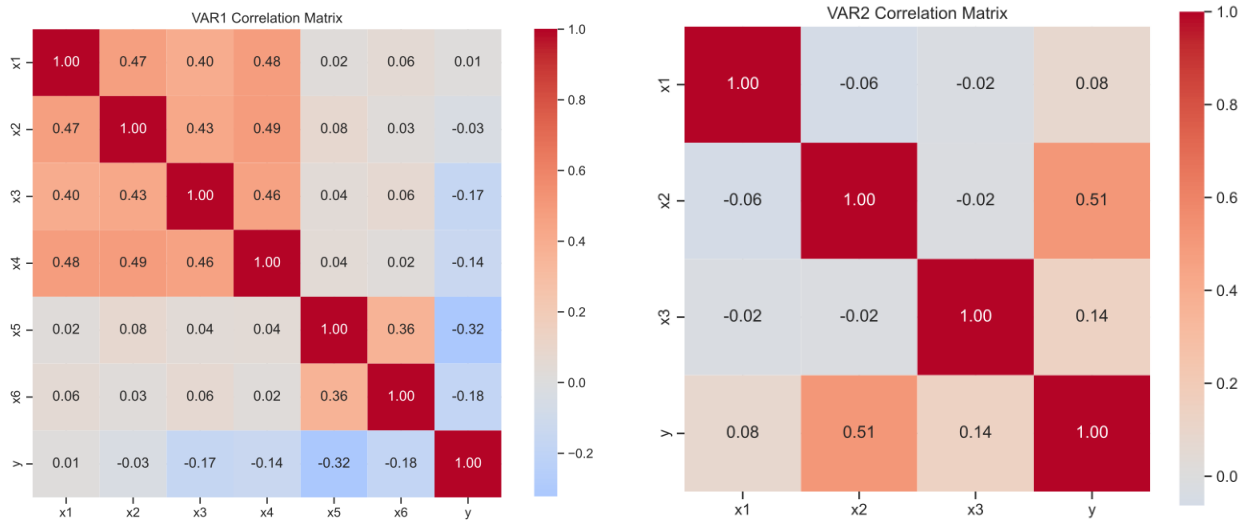


Figure 1: Correlation matrices for VAR1 (left) and VAR2 (right).

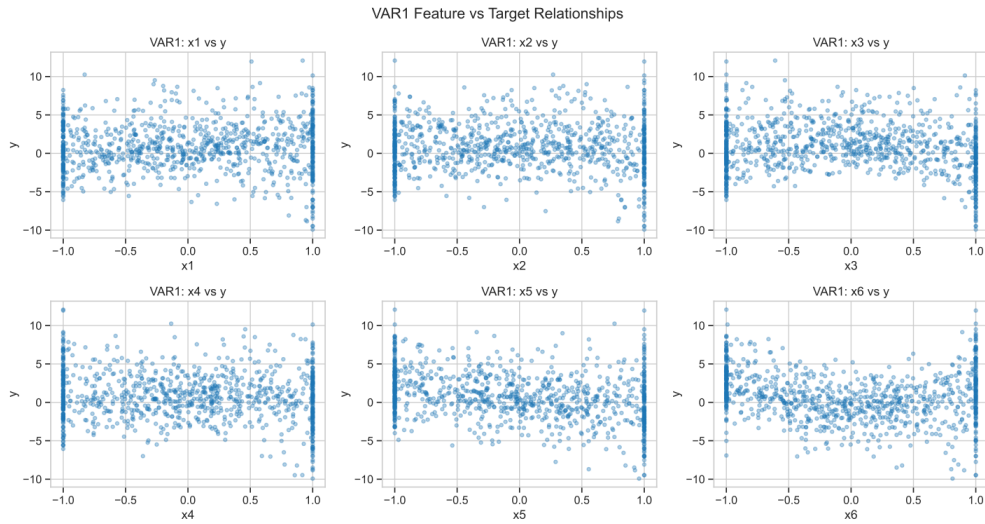


Figure 2: VAR1: each feature against y (training data).

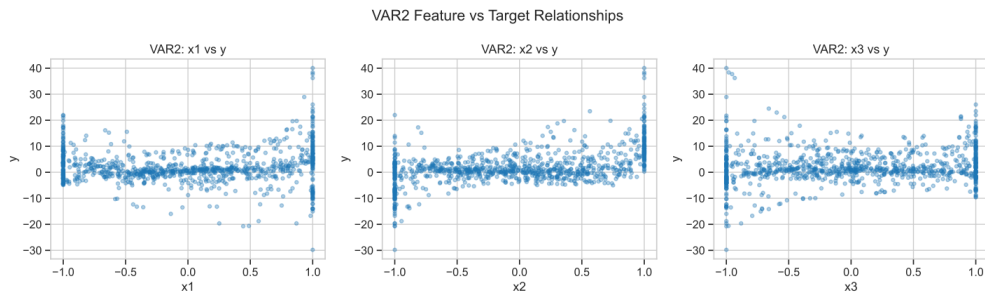


Figure 3: VAR2: each feature against y (training data).

for example $p(6, 4) = 209$, $p(6, 6) = 923$ and $p(3, 8) = 164$. The terms are then fitted with ordinary least squares (`LinearRegression`).

Other techniques used. Apart from the polynomial model itself, I used two things:

- **Standardisation** of the polynomial terms (`StandardScaler`). High-degree terms have very different sizes, and without scaling the least-squares solve becomes numerically unstable. With plain least squares, scaling does not change the model, it only makes the calculation stable.
- **5-fold cross-validation** (shuffled, `random_state=42`). The 1000 training rows are split into 5 folds, so each round trains on 800 rows and validates on 200. Comparing training error with validation error shows underfitting (both high) and overfitting (training low, validation high). Scaling is done inside a `Pipeline`, so it is learned only from the training part of each fold.

Only polynomial regression. I followed the rule strictly. I did not use regularised models such as Ridge or Lasso, and no ensemble or other model types. The only things I tuned were the degree and the choice of features.

Search. First I tried every degree from 1 to 10 for VAR1 and 1 to 20 for VAR2 (the maximum degrees mentioned in the problem statements) using all features. Then I tried every non-empty subset of features with every degree: $63 \times 10 = 630$ configurations for VAR1 and $7 \times 20 = 140$ for VAR2. I picked the combination with the lowest mean validation MSE and also looked at neighbouring degrees to check that the choice is not a lucky spike.

4 Choosing the Degree

Table 2: 5-fold cross-validation results with all features (VAR2 degrees 11 to 20 left out: validation MSE rises from 0.52 at degree 11 and becomes extremely large and numerically unstable for degrees 15 to 20).

degree	VAR1 (6 features)			VAR2 (3 features)		
	train MSE	val MSE	val R^2	train MSE	val MSE	val R^2
1	8.664	8.847	0.136	30.827	31.298	0.284
2	2.886	3.119	0.692	18.770	19.692	0.547
3	0.767	0.959	0.905	9.982	11.008	0.748
4	0.358	0.835	0.919	3.343	3.981	0.908
5	0.113	1.651	0.838	1.247	1.662	0.962
6	≈ 0	65.360	-5.348	0.454	0.647	0.985
7	≈ 0	6.989	0.320	0.230	0.338	0.992
8	≈ 0	4.312	0.585	0.164	0.295	0.993
9	≈ 0	3.923	0.618	0.144	0.306	0.993
10	≈ 0	4.347	0.582	0.125	0.383	0.991

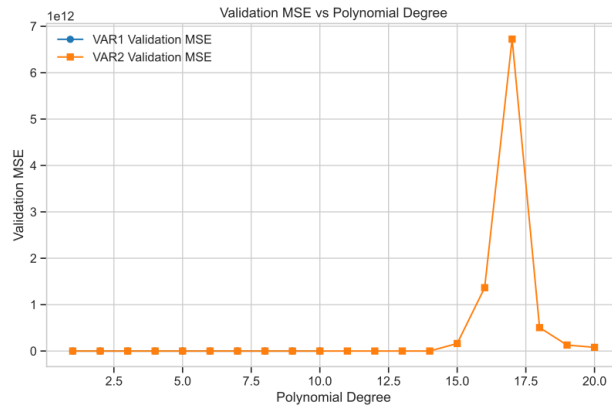


Figure 4: Validation MSE against polynomial degree for VAR1 and VAR2 (linear scale). The VAR2 error above degree 14 is so large that the other values look flat at this scale, so Table 2 gives the exact numbers for the useful range.

VAR1. Validation error keeps falling until degree 4 (MSE 0.835, $R^2 = 0.919$) and then goes up. At degree 6 the model has 923 terms, which is more than the 800 training rows in each fold, so it simply memorises the training data (training MSE is zero) and the validation MSE jumps to 65.4. Degrees 7 to 10 stay in this memorising regime and do poorly on unseen data. Degree 3 is close behind (0.959), but degree 4 is better, so I chose degree 4.

VAR2. Validation error drops smoothly to its minimum at degree 8 (MSE 0.295, $R^2 = 0.993$). Degrees 7 and 9 are close (0.338 and 0.306), so the choice is stable. This also fits the problem description, which says the heat map comes from complex geological structures: a fairly high degree is what I would expect, and the data agrees. Beyond degree 11 the model overfits quickly. At degrees 15 to 20 the validation error becomes extremely large and numerically unstable, which points to severe overfitting and ill-conditioning, and these models perform far worse than the selected degree-8 model.

Feature subsets. The best-performing configurations used the full feature set. For VAR1 the best model without one feature (dropping x_4 , degree 3) has a validation MSE of 1.798, more than twice that of the full model. For VAR2 the ten best configurations all use all three coordinates.

Table 3: Best results from the search over feature subsets and degrees.

Problem	Features	Degree	val MSE	val R^2
VAR1	$x_1, \dots, x_6$	4	0.835	0.919
VAR1	$x_1, \dots, x_6$	3	0.959	0.905
VAR1	x_1, x_2, x_3, x_5, x_6	3	1.798	0.822
VAR2	x_1, x_2, x_3	8	0.295	0.993
VAR2	x_1, x_2, x_3	9	0.306	0.993
VAR2	x_1, x_2, x_3	7	0.338	0.992

5 Final Models and Checks

I refitted the chosen pipelines on all 1000 training rows and used them to predict the 1000 test rows of each problem. The cross-validation scores in Table 2 are my estimate of out-of-sample performance. The true test performance is not known, because the test labels are hidden. The numbers in Table 4 are measured on the training data itself, so they are naturally a bit better.

Table 4: Fit and residuals of the final models on the training data.

	MSE	RMSE	R^2	mean res.	median res.	min res.	max res.
VAR1 (degree 4)	0.389	0.624	0.962	0.000	-0.008	-2.048	2.415
VAR2 (degree 8)	0.174	0.417	0.996	0.000	-0.001	-1.455	1.731

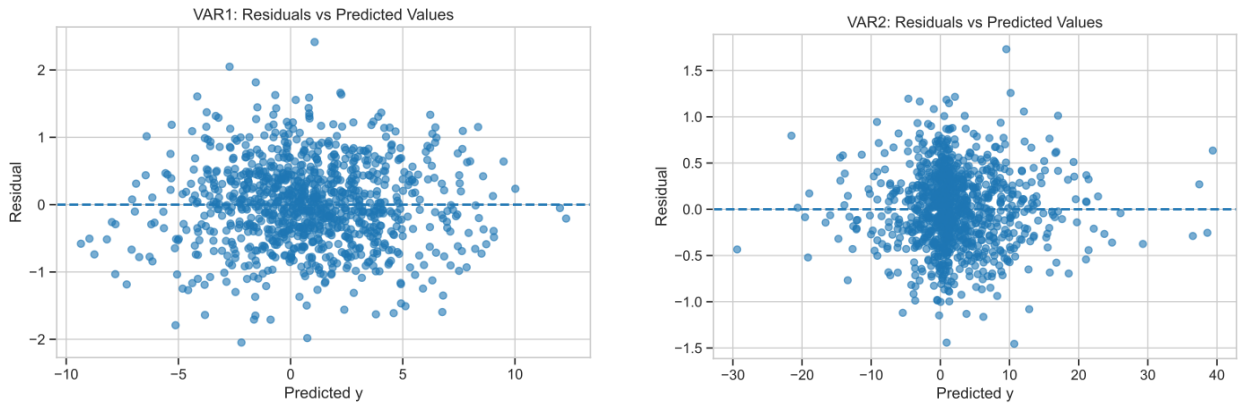


Figure 5: Residuals against predicted values on the training data: VAR1 (left), VAR2 (right).

The residual plots show no obvious curvature or funnel-shaped pattern, and the residuals are centred around zero. The Q-Q plots (Figure 6) follow a straight line closely, with only a few points off the line in the tails.

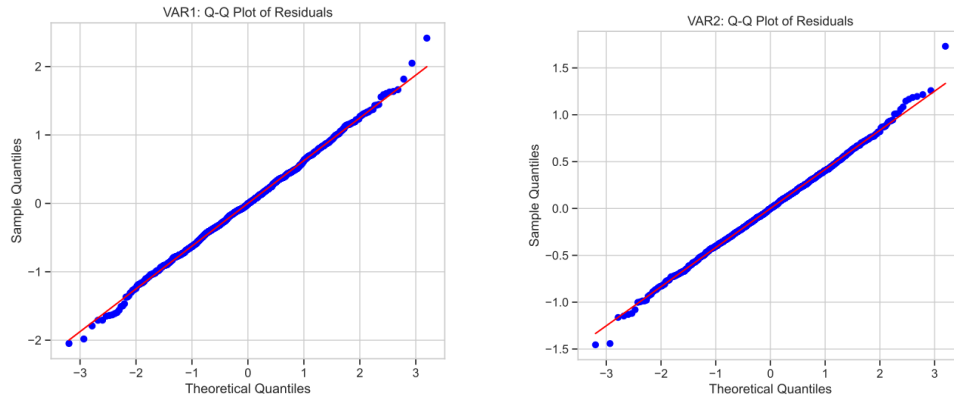


Figure 6: Q-Q plots of the training residuals: VAR1 (left), VAR2 (right).

6 Limitations

- **VAR1 test inputs are different.** Cross-validation assumes the test data looks like the training data. For VAR1 the test points are pushed towards the edges. The cross-validation estimate may not fully represent performance on the hidden test set. Eight of the 1000 VAR1 predictions fall outside the training range of y (-9.92 to 12.07), and the largest is 16.34 . None of the VAR2 predictions do. The higher-degree VAR1 models also performed poorly in cross-validation, which supports retaining the lower degree-4 model.
- **Slightly optimistic winner.** The same five folds were used to compare 770 configurations, so the best score is a little optimistic. The chosen degrees are supported by the neighbouring-degree results in Table 2, so the selection is not based on a single isolated score.
- **One random split.** All results use a single shuffle (seed 42).

7 Code and Files

All the code (data checks, plots, cross-validated search, final training and prediction) is in the notebook `ML_Assignment.ipynb`. It reads the four personalised training and test datasets from the `Dataset/` directory and writes the prediction files `IMT2023555_pred_var1.csv` and `IMT2023555_pred_var2.csv`. Each prediction file has 1000 rows in the same order as the test file, with one column y and no missing values. As a last check, I refitted both models separately from the training files and got the same predictions.

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